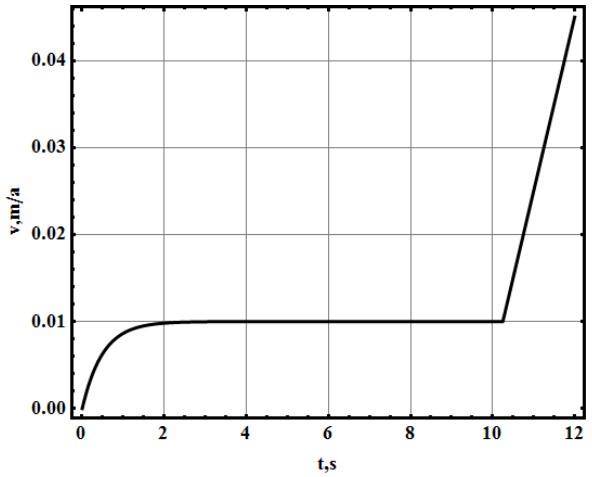
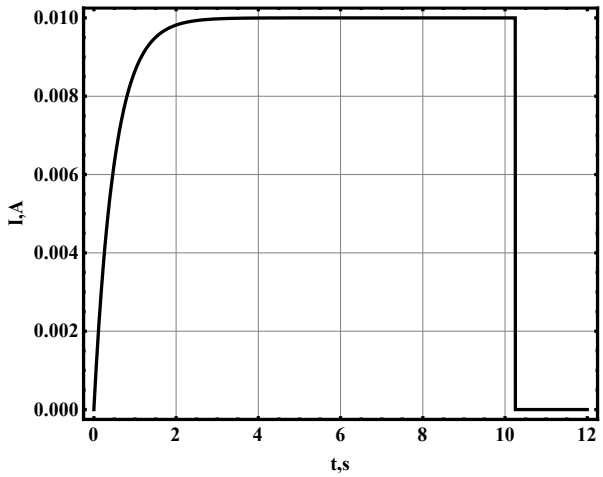


5. (3)

(6) . .

$$I(t)=\begin{cases}\dfrac{F}{Ba}\left[1-\exp\left(-\dfrac{B^2a^2}{mR}t\right)\right], & t<t_0\\ 0, & t\geq t_0\end{cases}. \qquad (12)$$

:



6. (1)

. :

$$\varepsilon_L=-L\frac{dI}{dt}. \qquad (13)$$

, :

$$Ba\mathrm{v}(t)-L\frac{dI}{dt}=0. \qquad (14)$$

$I=0,\ x=0$, : (13) (14)

$$I=\frac{Bax}{L}. \qquad (15)$$

:

$$F_A=BaI=\frac{B^2a^2x}{L}. \qquad (16)$$

:

$$m\frac{d^2x}{dt^2}=F-\frac{B^2a^2}{L}x. \qquad (17)$$

(17) :